

MSCF Probability: Homework Set #1

1. A health maintenance organization (HMO) concerned with the adverse financial effects of catastrophic medical expenses for its members enters into a reinsurance arrangement in which the reinsurer (i.e., the insurer providing insurance to the HMO) agrees to pay the amount of each plan member's annual hospital expenses that exceed \$100,000. Thus, for example, if the HMO had exactly two members and these two members had annual expenses of \$130,000 and \$70,000, respectively, then the reinsurer would pay the HMO \$30,000. Under the terms of the agreement, the HMO will be required to pay additional reinsurance premium if the fraction of the plan membership for which the reinsurance claim is filled exceeds 5%. This condition helps minimize any incentive the HMO may have to increase its enrollment of substandard risks after entering into the reinsurance arrangement.

The probability of catastrophic claim for any given member clearly depends on many factors, one of which is the age of the member. On the basis of historical data, the plan estimates that 10% of members over the age of 65 will incur hospital expenses exceeding \$100,000, 4% of members between the ages of 40 and 65 will incur hospital expenses exceeding \$100,000, and 3% of members under the age of 40 will incur hospital expenses exceeding \$100,000.

20% Twenty percent of the HMO's members are currently over the age of 65. If the HMO wishes to maintain this level of members over the age of 65 (perhaps for competitive or regulatory reasons) and wishes to avoid exceeding the 5% trigger point in the reinsurance agreement, how large must the HMO's under-40 population be?

Hint: Let C be the event that a member has hospital expenses exceeding \$100,000, use the Law of Total Probability and solve $P(C) \leq .05$.

2. A box contains three coins. Two of these are fairly unusual coins: one has heads on both sides, one has tails on both sides. The other is a fair coin. (one head and one tail, equally likely). You may assume each coin, when tossed, is equally likely to come up on either of its two faces.

A coin is chosen at random and tossed once; it comes up heads. What is the probability that the coin chosen is the fair coin?

(solution on the second page).

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1. Let E be the event that a member has hospital expenses exceeding \$10,000.

A : {member over age 65}.

B : { " over age 45 below 65 }

C : { " below age 45 }.

$$P(A) = 0.2,$$

$$P(B) = 0.8 - P(C),$$

$$P(C).$$

$$\begin{aligned} P(E) &= P(E|A)P(A) + P(E|B)P(B) + P(E|C)P(C) \\ &= 0.1 \times 0.2 + 0.04(0.8 - P(C)) + 0.03 \cdot P(C) \\ &= 0.052 - 0.01P(C) \leq 0.05. \end{aligned}$$

$$\therefore P(C) \geq 0.2.$$

$\Rightarrow$ The proportion of member under 45 must at least be 20%

2. A : {the coin chosen is a fair coin}.

$$P(A) = 1/3.$$

B : {toss once and comes up a head}.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B|A) \cdot P(A) + P(B|A^c) \cdot P(A^c)}$$

$$= \frac{1/2 \cdot 1/3}{1/2 \cdot 1/3 + 1 \cdot 1/3 + 0 \cdot 1/3}$$

$$= \frac{1/6}{1/6 + 1/3}$$

$$= \boxed{1/3}.$$